

PostgreSQL Replication Lab — Master → Replica

Vai trò / Role	Hostname	IP	Ghi chú / Notes
Primary / Master	Post1	192.168.232.13	Nhận ghi · gửi WAL / accepts writes, ships WAL
Replica 1 / Standby	Post2	192.168.232.14	Chỉ đọc / read-only standby

Install Postgre

```
hungdg@PC-BNKB167:~$ sudo systemctl status postgresql
● postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: enabled)
   Active: active (exited) since Tue 2026-06-30 14:58:42 +07; 3min 28s ago
  Invocation: 9cefa9d9f8c04ff185dc2e9a861c475c
    Main PID: 2737 (code=exited, status=0/SUCCESS)
   Mem peak: 1.9M
      CPU: 5ms

Jun 30 14:58:42 PC-BNKB167 systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...
Jun 30 14:58:42 PC-BNKB167 systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.
```

Cho phép Replica kết nối (pg_hba.conf)

```
post1@post1:~$ echo "host    replication    all    192.168.232.14/24    md5" | sudo tee -a /etc/postgresql/16/main/pg_hba.conf
host    replication    all    192.168.232.14/24    md5
post1@post1:~$ sudo systemctl restart postgresql
```

Khởi động lại & 2.4 — Tạo user replication

```
post1@post1:~$ sudo systemctl restart postgresql@16-main
post1@post1:~$ pg_lsclusters
Ver Cluster Port Status Owner    Data directory          Log
file
16  main    5432 online postgres /var/lib/postgresql/16/main /var/log/postgresql/postgresql-16-main.log
post1@post1:~$ sudo -u postgres psql -c "CREATE ROLE replication_
user WITH REPLICATION PASSWORD '123' LOGIN;"
CREATE ROLE
post1@post1:~$ |
```

Thiết lập Replica · Set up replication

Dừng dịch vụ & dọn thư mục dữ liệu

```
post2@post2:~$ sudo rm -rf /var/lib/postgresql/16/main
post2@post2:~$ sudo mkdir -p /var/lib/postgresql/16/main
post2@post2:~$ sudo chown -R postgres:postgres /var/lib/postgresql/16/main
post2@post2:~$ sudo chmod 700 /var/lib/postgresql/16/main
post2@post2:~$ sudo ls -la /var/lib/postgresql/16/main/
total 8
drwx----- 2 postgres postgres 4096 Jun 30 07:32 .
drwxr-xr-x 3 postgres postgres 4096 Jun 30 07:32 ..
post2@post2:~$
```

Sao chép dữ liệu bằng pg_basebackup

```
post2@post2:~$ sudo -u postgres pg_basebackup -h 192.168.232.13 -D /var/lib/postgresql/16/main -U replication_user -v -P --wal-method=stream -R
Password:
pg_basebackup: initiating base backup, waiting for checkpoint to complete
pg_basebackup: checkpoint completed
pg_basebackup: write-ahead log start point: 0/2000028 on timeline 1
pg_basebackup: starting background WAL receiver
pg_basebackup: created temporary replication slot "pg_basebackup_1993"
23094/23094 kB (100%), 0/1 tablespace (...gresql/16/main/global/p
23094/23094 kB (100%), 1/1 tablespace
pg_basebackup: write-ahead log end point: 0/2000100
pg_basebackup: waiting for background process to finish streaming
...
pg_basebackup: syncing data to disk ...
pg_basebackup: renaming backup_manifest.tmp to backup_manifest
pg_basebackup: base backup completed
```

! lưu ý start máy replica rồi mới chạy select trên master

```
post2@post2:~$ sudo systemctl start postgresql@16-main
```

[illegible]

Test replication

Tạo dữ liệu trên Master

```
post1@post1:~$ sudo -u postgres psql -c "create database labdb;"
CREATE DATABASE
post1@post1:~$ sudo -u postgres psql -d labdb -c "
CREATE TABLE IF NOT EXISTS test_replication (
    id SERIAL PRIMARY KEY,
    message TEXT,
    created_at TIMESTAMP DEFAULT now()
);
INSERT INTO test_replication(message) VALUES ('Hello from master'
);
SELECT * FROM test_replication;
"
CREATE TABLE
INSERT 0 1
 id |          message          |          created_at
----+-----
```

id	message	created_at
1	Hello from master	2026-06-30 07:44:07.277794

```
(1 row)
```

```
post1@post1:~$ |
```

Đọc dữ liệu trên Replica

```
post2@post2:~$ sudo -u postgres psql -d labdb -c "SELECT * FROM test_replication;"
 id |          message          |          created_at
----+-----
```

id	message	created_at
1	Hello from master	2026-06-30 07:44:07.277794

```
(1 row)
```

Mở kết nối từ xa

```
post1@post1:~$ sudo sed -i "s/#listen_addresses = 'localhost'/listen_addresses = '*' /g" /etc/postgresql/16/main/postgresql.conf
post1@post1:~$ echo "host    all    all    0.0.0.0/0    md5" |
sudo tee -a /etc/postgresql/16/main/pg_hba.conf
host    all    all    0.0.0.0/0    md5
post1@post1:~$ sudo systemctl restart postgresql
post1@post1:~$ sudo ss -tuln | grep 5432
tcp     LISTEN 0      200                0.0.0.0:5432        0.0.0.0:
0:*
tcp     LISTEN 0      200                [::]:5432          [::]:
]:*
```

Tạo user cho DBeaver

```
post1@post1:~$ sudo -u postgres psql
psql (16.14 (Ubuntu 16.14-0ubuntu0.24.04.1))
Type "help" for help.

postgres=# CREATE USER dbeaver_user WITH PASSWORD '123';
CREATE ROLE
postgres=# GRANT ALL PRIVILEGES ON DATABASE labdb TO dbeaver_user;
GRANT
postgres=# GRANT ALL ON SCHEMA public TO dbeaver_user;
GRANT
postgres=# GRANT ALL PRIVILEGES ON ALL TABLES IN SCHEMA public TO dbeaver_user;
GRANT
postgres=# ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT ALL ON TABLES TO dbeaver_user;
ALTER DEFAULT PRIVILEGES
postgres=#
```

Kiểm tra kết nối từ máy Client

```
post2@post2:~$ nc -zv 192.168.232.13 5432
Connection to 192.168.232.13 5432 port [tcp/postgresql] succeeded
!
```